Optimal Treatment

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Optimal Policies

- Optimal policies based on structural model: *e.g.* MPEC

$$\begin{aligned} &\max_{X} \mathcal{O}\left(Y, X, \theta, \epsilon\right) \\ &\text{s.t. } G\left(Y, X, \theta, \epsilon\right) = 0 \end{aligned} \tag{1}$$

- A full structural model has to be specified and estimated

- Can we characterize optimal policies based on reduced-form estimates of $X \to Y$?

- Assuncao et al. (RES, 2023) "Optimal Environmental Targeting in the Amazon Rainforest"

Institutional Background and Regulations

- Regulating deforestation in the Brazilian Amazon
- Priority List
  - Launched in 2008
  - Municipalities placed on the Priority List were subject to more intense environmental monitoring and law enforcement
  - Selection criteria: the exact rules are not in the public domain
    - Total deforestated area
    - Total deforested area over the past 3 years
    - The increase in the deforestation rate in at least 3 of the past 5 years

Data

- Municipality-year panel over 2006-2010
  - Treated: 35 municipalities that entered and remained on the list from 2008 to 2010 inclusive
  - Untreated: municipalities that did not enter the list before 2010
- Variables:
  - Deforestation: incremental; cumulative; forested area remaining
  - Factors affecting deforestation: rainfall; temperature; protected area; prices of beef and crops; local GDP; soil quality; distance to the nearest port; crop area; the number of cattle

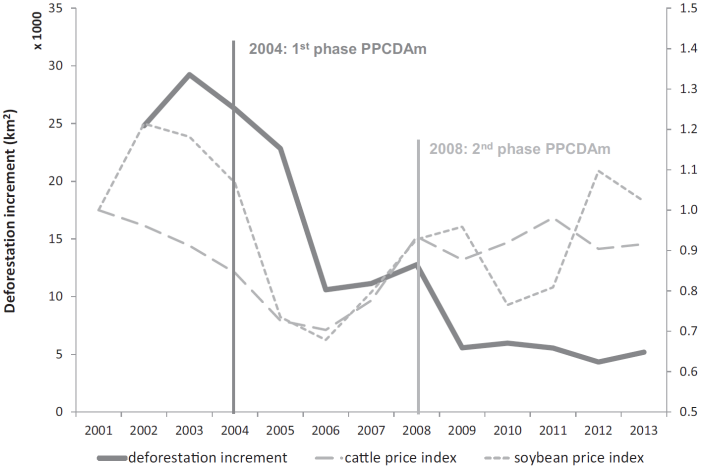


FIGURE 1

Incremental deforestation (leftmost vertical axis), key policy changes, and agricultural commodity prices (rightmost axis), by year.

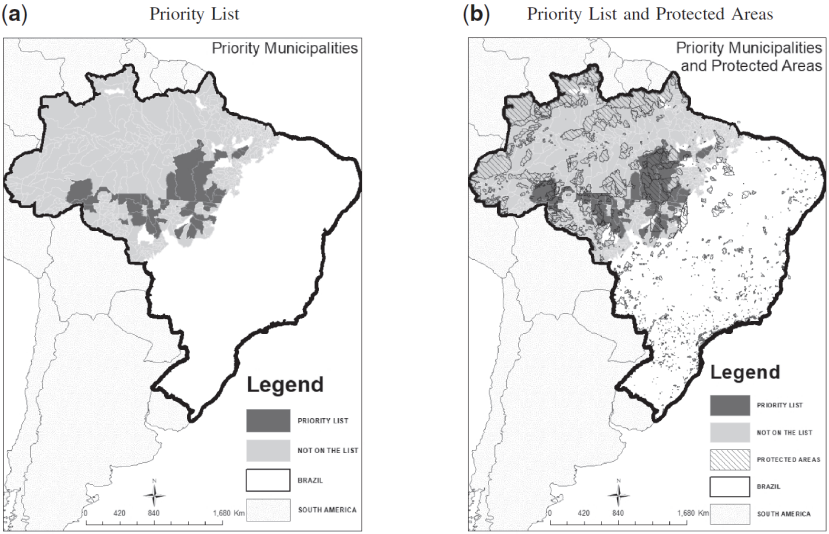


FIGURE 2

Location of Priority List and protected areas.

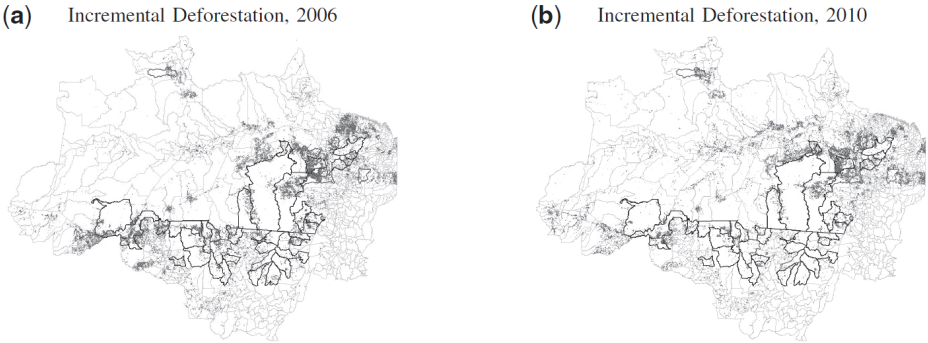


FIGURE 3

Map of deforestation in 2006 and 2010 (with priority municipalities overlaid).

Selection onto the Priority List

$$G_{mt} = g\left(Z_{mt-1}\right), \quad G_{mt} \in \{0,1\}, \quad Z_{mt-1} \equiv \left(Z^1_{mt-1}, Z^2_{mt-1}, Z^3_{mt-1}\right) \tag{2}$$

1. Z^1_{mt-1}: total amount of forested land cleared in municipality m from its inception up to $t-1$
2. Z^2_{mt-1}: amount of forested land cleared in m in the 3-year period ending in $t-1$
3. Z^3_{mt-1}: whether m experienced year-on-year growth in the new deforestation at least three times in the 5-year period ending with $t-1$

Selection onto the Priority List

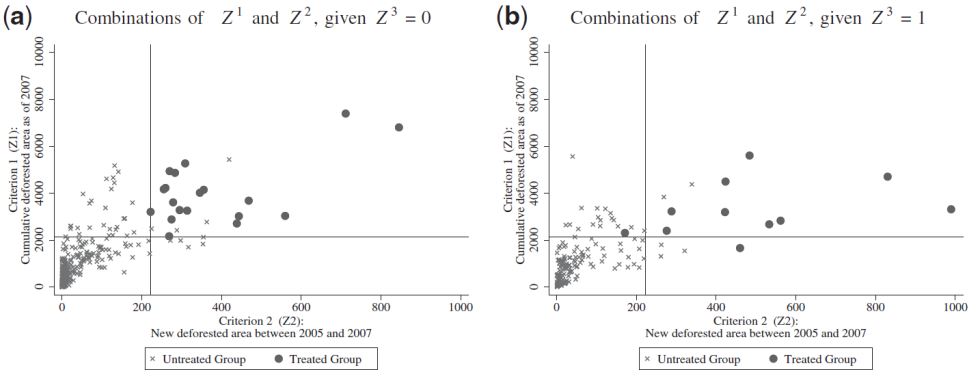


FIGURE 4

Selection onto priority list in 2008—combinations of criteria variables (Z_{mt-1}).

Selection onto the Priority List

- Close to a threshold rule. Pre-determined thresholds
  - Z^1_{mt-1}: 2137 km^2
  - Z^2_{mt-1}: 222 km^2
  - Z^3_{mt-1}: not important
- Using these inferred thresholds, replicate the actual 2008 assignments with 98% accuracy
- Local political influence are unlikely to affect the Priority List's initial composition

Changes-in-Changes

- $Y_{mt} \equiv D_{mt}/A_{mt}$: the share of newly deforested area

$$Y_{mt} = (1 - G_m) \times Y^0_{mt} + G_m \times Y^1_{mt} \tag{3}$$

- The potential share of deforestation:

$$Y^j_{mt} = h^j\left(X_{mt}, U_{mt}, t\right) \tag{4}$$

  - X_{mt}: a municipality-level vector of observed factors, including prices and agro-climatic conditions
  - U_{mt}: a municipality-level unobserved term that can incorporate municipality fixed effects

Changes-in-Changes

- Assumption #1: Monotonicity: The function $h^j(x, u, t)$, for $j \in \{0, 1\}$, are strictly increasing in u—the same restriction applies to the standard DID model

- Assumption #2: Time invariance within groups: Conditional on each group G and on observable X, the unobservable U has an identical distribution over time—similar to the parallel trends assumption in the standard DID model

Changes-in-Changes

- $F_{Y^j_{gt}}$: the conditional distribution function of the potential share of deforestation Y^j_{mt} given $G = g$ and $X = x$

- $F^{-1}_{Y^j_{gt}}(q)$: any quantile $q \in [0, 1]$

- Athey and Imbens (2006) show that under Assumption 1 and 2, the counterfactual distribution of Y^0_{1mt+1} is given by

$$F_{Y^0_{1t+1}}(y) = F_{Y_{1t}}\left(F^{-1}_{Y_{0t}}\left(F_{Y_{0t+1}}(y)\right)\right). \tag{5}$$

  - $F_{Y_{1t}}$: the distribution of deforestation shares for the same group but prior to treatment

  - $F_{Y_{0t}}$ and $F_{Y_{0t+1}}$: the distributions of the share of deforestation for the control group both before and after treatment

Changes-in-Changes

- The counterfactual distribution of the treated group, $F_{Y^0_{1t+1}}$, is identified only on the support of the control group $\mathbb{Y}_{0t+1}$

- Assumption #3 Support: $\mathbb{Y}^j_{gt} = Y_{gt}$ for $j, g = 0, 1$ and for any t—while the policy interventions may affect the distribution of deforestation shares, it does not affect the support of the distribution

- Assumption #4 Semi-parametric model: The potential share of newly deforested area, Y^j_{mt}, for $j \in \{0, 1\}$, in a municipality m at t is given by

$$Y^j_{mt} = \frac{\exp\left[X'_{mt}\beta + V^j_{mt}\right]}{1 + \exp\left[X'_{mt}\beta + V^j_{mt}\right]}, \tag{6}$$

where $V^j_{mt} = v^j(U_{mt}, t)$, with $v^j(u, t)$ satisfying Assumption 1; U_{mt} is independent of X_{mt} given G_m; V^j_{mt} satisfy Assumption 3

Changes-in-Changes

- Logistic function $\varphi(x, \nu) = \exp(x'\beta + \nu) / (1 + \exp(x'\beta + \nu))$. The potential share of new deforestation is:

$$Y_{mt}^{j} = \varphi\left(X_{mt}, V_{mt}^{j}\right). \tag{7}$$

- Average treatment effect

$$E\left[D_{mt}^{j} | X_{mt}, A_{mt}, G_m = g\right] = \int \left[\varphi\left(X_{mt,\nu}\right) \times A_{mt}\right] dF_{V_{gt}^{j}}(v), \tag{8}$$

where the distribution $F_{V_{gt}^{j}}$ is either observed or is obtained from the CIC model

Changes-in-Changes

TABLE 2

Average and cumulative treatment effects, without spillovers

Average treatment effects: deforestation (km^2)			
	ATT	ATU	ATE
2009	−24.49	[−4.04, −3.67]	[−5.50, −5.15]
	(−27.64, −21.34)	(−4.16, −3.58)	(−5.65, −5.04)
2010	−57.97	[−6.84, −5.95]	[−10.49, −9.66]
	(−62.95, −52.99)	(−6.97, −5.85)	(−10.66, −9.52)
Cumulative treatment effects, 2009–10			
	Deforestation (km^2)	Carbon Emissions (millions tC)	Value (billions U$)
Total effects	−2886	−34.60	2.54
	(−3145, −2627)	(−37.96, −31.24)	(2.29, 2.78)

Notes: 95% confidence intervals are in parentheses. For ATT and the cumulative effects, the confidence intervals are computed based on the standard i.i.d. non-parametric bootstrap, where the i.i.d. resampling occurs in the cross-sectional dimension. For ATU and ATE, they are based on Imbens and Manski (2004). We implemented 500 bootstrap replications. Deforestation is measured in square kilometres. Emissions are measured in millions of tons of carbon. Values are measured in billion US$, assuming a social cost of carbon of US$ 20/tCO2. The calculations use the fact that 1 tC = (44/12) tCO2.

Changes-in-Changes

TABLE 3

Average and cumulative treatment effects, with spillovers

Average treatment effects: deforestation (km^2)				
	ATT	ATU	ATS	ATE
2009	−28.30	[−3.91, −3.46]	[−11.82, −11.78]	[−6.22, −5.83]
	(−31.63, −24.96)	(−4.01, −3.39)	(−13.78, −9.91)	(−6.36, −5.70)
2010	−58.62	[−6.19, −5.66]	[−20.27, −18.07]	[−10.94, −10.33]
	(−63.52, −53.72)	(−6.30, −5.58)	(−22.10, −16.02)	(−11.11, −10.18)

Cumulative treatment effects, 2009–10			
	Deforestation (km^2)	Carbon emissions (millions tC)	Value (billions U$)
Direct effects	−3042	−36.39	2.67
	(−3304, −2780)	(−39.83, −32.95)	(2.42, 2.92)
Indirect effects	−1120	−13.11	0.96
	(−1249, −992)	(−14.70, −11.53)	(0.85, 1.08)

Notes: 95% confidence intervals are in parentheses. For ATT and the cumulative direct effects, the intervals are computed based on the standard i.i.d. non-parametric bootstrap, where the i.i.d. resampling occurs in the cross-sectional dimension. For ATU, ATS, ATE, and the cumulative indirect effects, they are based on Imbens and Manski (2004). We implemented 500 bootstrap replications. Deforestation is measured in square kilometres. Emissions are measured in millions of tons of carbon. Values are measured in billion US$, assuming a social cost of carbon of US$ 20/tCO2. The calculation uses the fact that 1 tC = (44/12) tCO2.

Optimal Treatment

- The expected deforestation under intervention j: $\mathcal{D}^j_{mt} \equiv E\left[D^j_{mt} | X_{mt}, A_{mt}, G_m\right]$

$$\min_{\phi_t \in [0,1]^M} \sum_{m=1}^{M} \left[\phi_{mt} \mathcal{D}^1_{mt} + (1 - \phi_{mt}) \mathcal{D}^0_{mt}\right]. \tag{9}$$

- Constraints:
  - Version #1: $\sum_{m=1}^{M} s_m \times \phi_{mt} \leq \bar{S}$
  - Version #2: $\sum_{m=1}^{M} \phi_{mt} \leq \bar{M}$.

Optimal Treatment

TABLE 5

Percent correctly predicted, ex post optimal list

No spillovers case						
Constraint:	Total area			Number of municipalities		
	Optimal			Optimal		
	0	1		0	1	
Observed			Percent correct			Percent correct
0	383	72	84.18	439	16	96.48
1	9	26	74.29	16	19	54.29
		Overall	83.47		*Overall*	93.47
Spillovers case						
Constraint:	Total area			Number of municipalities		
	Optimal			Optimal		
	0	1		0	1	
Observed			Percent correct			Percent correct
0	336	119	73.85	435	20	95.60
1	20	15	42.86	20	15	42.86
		Overall	71.63		Overall	91.84

Notes: Authors' calculations.

Optimal Treatment

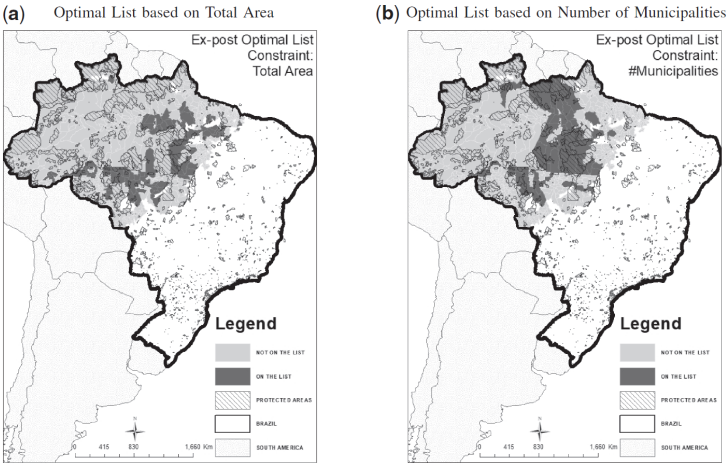


FIGURE 5

Location of *ex post* optimal lists without spillovers and protected areas.

Optimal Treatment

TABLE 6
Comparing ex post optimal, priority, and randomly selected lists

No spillovers case						
Constraint:	Total area		Number of municipalities			
	Observed vs. optimal		Observed vs. optimal		Random vs. optimal	
	Ratio	Value	Ratio	Value	Ratio	Value
Total deforestation	1.06	–	1.06	–	1.22	–
Total carbon emissions	1.05	622	1.08	975	1.26	3,094
Spillovers case						
Constraint:	Total area		Number of municipalities			
	Observed vs. optimal		Observed vs. optimal		Random vs. optimal	
	Ratio	Value	Ratio	Value	Ratio	Value
Total deforestation	1.12	–	1.10	-	1.26	-
Total carbon emissions	1.11	1,200	1.11	1,224	1.30	3,256

Notes: "Ratio" divides total deforestation (total emissions) evaluated at the observed list by the *ex post* optimal total deforestation (total emissions). "Value" takes their difference. Values are measured in million US$, assuming a social cost of carbon = US$ 20/tCO2.

Optimal Treatment

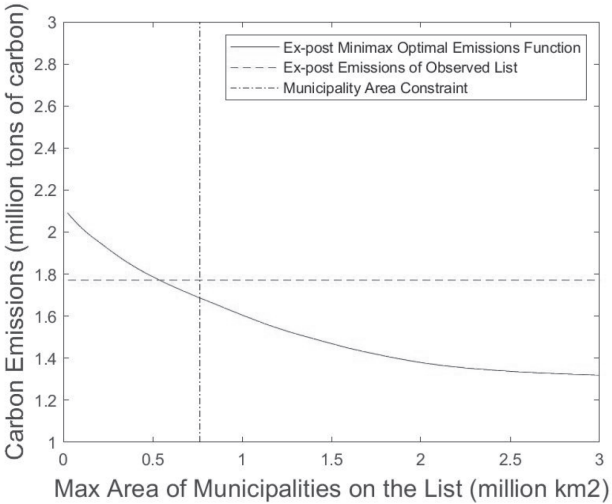
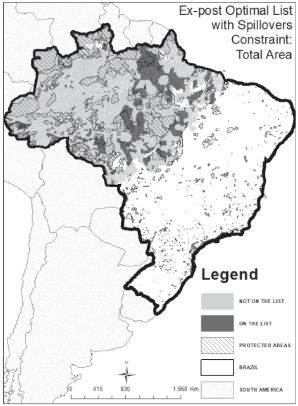


FIGURE 6

Ex post minimax carbon emissions, varying the "total area" constraint

Optimal Treatment

(a) Spillover Optimal List based on Total Area



(b) Spillover Optimal List based on Number of Municipalities

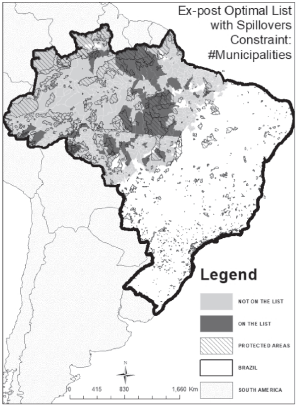


FIGURE 7

Location of *ex post* optimal lists with spillover effects.

Summary

- Estimating potential outcomes for each observation i: Y_i^0 and Y_i^1
- Objective:

$$\max_{\phi_i \in [0,1]} \sum_{i=1}^{N} \omega_i \times \left[\phi_i Y_i^1 + (1 - \phi_i) Y_i^0 \right]. \tag{10}$$

- Intervention cost:

$$\sum_{i=1}^{N} s_i \times \phi_i \leq \bar{S}. \tag{11}$$

- Ex-post evaluation
- *e.g.* Re-evaluate the impacts of China shocks and optimally assign China shocks across U.S. regions?